GROUP 23

PROJECT CYCLE 3 SUMMARY

Two-Proportion Inference and Descriptive Extension

Cycle 3: Gender and Current Alcohol Use | Dataset: YRBS 2007 | Members: 周朝賢, 吳佳虹 | $\alpha = 0.05$

1. RESEARCH QUESTION

1. Is the proportion of current alcohol use different between male and female students?

2. Does the pattern look different when we compare male and female students within each grade level?

2. DATA

Dataset: YRBS 2007

Variables:

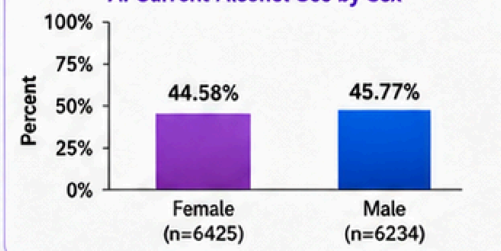
- WhatsYourSex (group)
- CurrentAlcoholUse (binary response)
- InWhatGradeAreYou (extension)

Sample size:

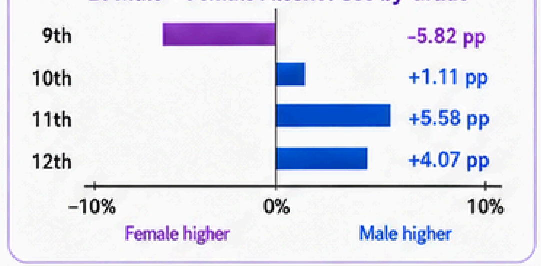
- Female: n = 6,425
- Male: n = 6,234
- Total: n = 12,659

3. EDA HIGHLIGHTS

- ✓ Current alcohol use is very similar overall for female and male students. Female: 44.58%; Male: 45.77%.
- ✓ The grade-level pattern varies: female is slightly higher in 9th grade, while male is higher in grades 10–12.



A. Current Alcohol Use by Sex



B. Male – Female Alcohol Use by Grade

4. METHOD

Two-proportion z-test

$H_0: p_{male} = p_{female}$

95% CONFIDENCE INTERVAL

for $p_{male} - p_{female}$

EXTENSION: DESCRIPTIVE GRADE COMPARISON

compare male and female proportions within each grade

5. KEY RESULTS

A. MAIN RESULT

- Female proportion = 44.58%
- Male proportion = 45.77%
- Difference (Male – Female) = +1.19 percentage points
- 95% CI = -0.54% to 2.92%
- z = 1.3442
- p = 0.1789

Conclusion: Fail to reject H_0 ($p > 0.05$).

No significant overall difference between male and female students.

B. EXTENSION (GRADE LEVEL)

Grade	Male – Female (pp)	Interpretation
9th grade	-5.82 pp	female higher
10th grade	+1.11 pp	male higher
11th grade	+5.58 pp	male higher
12th grade	+4.07 pp	male higher

Descriptive takeway: the overall result can hide grade-level variation.

6. CONCLUSION / KEY TAKEAWAY

- At $\alpha = 0.05$, we fail to reject H_0.
- There is not enough statistical evidence to conclude that current alcohol use differs between male and female students overall.
- However, the *extension suggests that the pattern is not identical across grade levels, so grade should be considered when interpreting the result.*

7. NOTE / LIMITATION

- Observational survey data: interpret results as association, not causation.
- The extension is descriptive and does not include separate significance tests within each grade.

Group 23 » Overall, the inferential result shows no significant overall difference, while the descriptive extension reveals meaningful grade-level variation.